

Directions for Use

Please Read Carefully!

Carvedilol HEXAL® 6.25/12.5/25 mg Tablet

Content

Each Carvedilol HEXAL® 6.25mg, Carvedilol HEXAL® 12.5mg and Carvedilol HEXAL® 25mg Tablet contains 6.25mg, 12.5mg and 25mg carvedilol respectively for the purpose of oral administration.

Dosage Form

Tablet

Description

Carvedilol HEXAL® 6.25mg Tablet: Yellow, round, convex, bisected tablet with a pressure-sensitive scoring notch, encoded C2.

Carvedilol HEXAL® 12.5mg Tablet: Red/brown, round, convex, bisected tablet with a pressure-sensitive scoring notch, encoded C3.

Carvedilol HEXAL® 25mg Tablet: White, round, convex, bisected tablet with a pressure-sensitive scoring notch, encoded C4.

Pharmacodynamics

Carvedilol is a vasodilative non-selective beta-blocker which reduces the peripheral vascular resistance by selective α_1 -receptor blockade and suppresses the renin-angiotensin system through non-selective beta-blockade. Plasma renin activity is reduced, and fluid retention is rare.

Carvedilol has no intrinsic sympathomimetic activity (ISA). Like propranolol, it has membrane-stabilizing properties.

Carvedilol is a racemate of two stereoisomers. Both enantiomers were found to have α_1 -adrenergic blocking activity. Non-selective β_1 - and β_2 -adrenoceptor blockade is attributed mainly to the S (-) enantiomer.

Carvedilol is a potent antioxidant, a scavenger of reactive oxygen radicals and anti-proliferative agent. The antioxidant properties of carvedilol and its metabolites have been demonstrated in *in vitro* and *in vivo* animal studies and *in vitro* in a number of human cell types.

In hypertensive patients, a reduction in blood pressure is not associated with a concomitant increase in peripheral resistance, as observed with pure beta-blocking agents. Heart rate is slightly decreased. Stroke volume remains unchanged. Renal blood flow and renal function remain normal, as does peripheral blood flow, therefore, cold extremities, often observed with beta-blockers, are only rarely seen. In hypertensive patients, carvedilol increases the plasma concentration of norepinephrine.

In prolonged treatment of patients with angina pectoris, carvedilol has been seen to have an anti-ischaemic effect and to alleviate pain. Haemodynamic studies demonstrated a reduction in ventricular pre- and after-load. In patients with left-ventricular dysfunction or congestive heart failure, carvedilol has a favourable effect on haemodynamic parameters and left-ventricular ejection fraction and dimensions.

Carvedilol has no negative effect on the serum lipid levels or electrolytes. The ratio of HDL (high-density lipoproteins) and LDL (low-density lipoproteins) remains normal.

Pharmacokinetics

Absorption

Following oral administration, carvedilol is rapidly absorbed. Carvedilol is a substrate of the intestinal efflux transporter P-glycoprotein which plays a major role in the bioavailability of certain drugs. In healthy volunteers the maximum serum concentration is reached after approximately one hour. The absolute bioavailability of carvedilol in humans is approximately 25%.

Distribution

Carvedilol is a highly lipophilic compound, approximately 98% to 99% bound to plasma proteins. The distribution volume is approx. 2 l/kg.

Metabolism

In humans, carvedilol is extensively metabolized in the liver via oxidation and conjugation into a variety of metabolites that are eliminated mainly in the bile. The first-pass effect after oral administration amounts to about 60-75%. Enterohepatic circulation of the parent substance has been shown in animals. The oxidative metabolism of carvedilol is stereoselective. The R-enantiomer is predominantly metabolized by CYP2D6 and CYP1A2, while the S-enantiomer is mainly metabolised by CYP2C9 and to a lesser extent by CYP2D6. Other CYP450 isoenzymes involved in the metabolism of carvedilol include CYP3A4, CYP2E1 and CYP2C19.

The maximal plasma concentration of R-carvedilol is approximately 2 fold higher than that of S-carvedilol. The R-enantiomer is predominantly metabolised through hydroxylation. In slow metabolisers of CYP2D6 an increase of the plasma concentration of carvedilol mainly the R-enantiomer may occur, leading to an increase in the α -blocking activity. Demethylation and hydroxylation at the phenol ring produce 3 metabolites with β -adrenergic receptor blocking activity.

Based on preclinical studies, the 4'-hydroxyphenol metabolite is approximately 13 times more potent than carvedilol for β -blockade. Compared to carvedilol, the three active metabolites exhibit weak vasodilating activity. In humans, the concentrations of the three active metabolites are about 10 times lower than that of the parent substance. Two of the hydroxy-carbazole metabolites of carvedilol are extremely potent antioxidants, demonstrating a 30 to 80-fold greater potency than Carvedilol.

Elimination

The average elimination half-life of carvedilol is approximately 6 hours. Plasma clearance is approximately 500-700 ml/min. Elimination is mainly

via the bile and excretion mainly via the faeces. A minor part is eliminated renally in the form of various metabolites.

Pharmacokinetics in Special Populations

Patients with renal impairment

The autoregulatory blood supply is preserved and the glomerular filtration is unchanged during chronic treatment with carvedilol. In patients with hypertension and renal insufficiency, the area under plasma level-time curve, elimination half-life and maximum plasma concentration does not change significantly. Renal excretion of the unchanged drug decreases in the patients with renal insufficiency; however, changes in pharmacokinetic parameters are modest.

Several open studies have shown that carvedilol is an effective agent in patients with renal hypertension. The same is true in patients with chronic renal failure, or those on hemodialysis or after renal transplantation. Carvedilol causes a gradual reduction in blood pressure both on dialysis and non-dialysis days, and the blood pressure-lowering effects are comparable with those seen in patients with normal renal function. Carvedilol is not eliminated during dialysis because it does not cross the dialysis membrane, probably due to its high plasma protein binding. On the basis of results obtained in comparative trials on hemodialysed patients, it was concluded that carvedilol was more effective than calcium channel blockers and was better tolerated.

Patients with hepatic impairment

In patients with cirrhosis of the liver, the systemic availability of the drug is increased by up to 80% because of a reduction in the first-pass effect. Therefore, carvedilol is contraindicated in patients with clinically manifest liver dysfunction.

Patients with hepatic failure

In a study of liver cirrhosis patients, bioavailability of Carvedilol has been observed to be 4x greater, peak plasma levels 5x higher and volume of distribution 3x higher compared to healthy patients.

Patients with heart failure

The clearance of R- and S-carvedilol was significantly lower than previously estimated in healthy volunteers. These results suggested that the pharmacokinetics of R- and S-carvedilol is significantly altered by cardiac failure.

Geriatric use

Age has no statistically significant effect on the pharmacokinetics of carvedilol in hypertensive patients. A study in elderly hypertensive patients showed that there was no difference in the adverse event profile compared to younger patients. Another study which included elderly patients with coronary heart disease showed no difference in the adverse events reported vs those reported by younger patients.

Pediatric use

There is limited data available on pharmacokinetics in people younger than 18 years of age.

Diabetic patients

In hypertensive patients with non-insulin-dependent diabetes no influence of carvedilol on fasting or post-prandial blood glucose concentration, glycalated hemoglobin A1 or need for change of the dose of antidiabetic agents was found. In patients with non-insulin dependent diabetes, carvedilol had no statistically significant influence on the glucose tolerance test. In hypertensive non-diabetic patients with impaired insulin sensitivity (syndrome X) carvedilol improved the insulin sensitivity. The same results were found in hypertensive patients with non-insulin dependent diabetes.

Indications

- **Hypertension**
Carvedilol HEXAL is indicated primarily for the management of essential hypertension. It can be used alone or in combination with other antihypertensive agents (e.g. calcium channel blockers, diuretics).
- **Treatment of angina pectoris**
- **Chronic Heart Failure**
Unless a contraindication exists, carvedilol is indicated for the treatment of all patients with stable and symptomatic, mild, moderate and severe chronic heart failure of ischemic or non-ischemic etiology in combination with standard therapy (including ACE inhibitors and diuretics with or without digitalis).

Recommended Dose

Duration of treatment

Treatment with carvedilol is a long-term therapy. As with all β -blockers, treatment should not be stopped abruptly but rather gradually reduced at weekly intervals. This is particularly important in the case of patients with concomitant coronary heart disease.

Hypertension

The recommended dose for initiation of therapy is 12.5 mg once a day for the first two days. Thereafter, the recommended dosage is 25 mg once a day. If necessary, the dosage may subsequently be increased at intervals of at least two weeks to the recommended maximum daily dose of 50 mg, given once or twice daily.

Coronary Heart Disease

The recommended dose for initiation of therapy is 12.5 mg twice a day for the first two days. Thereafter the recommended dosage is 25 mg twice a day. If necessary, the dosage may thereafter be increased at intervals of at least two weeks, up to the recommended maximum daily dose of 100 mg

given in divided doses (twice daily).

Chronic Heart Failure

Dosage must be tailored to suit the individual and closely monitored by a physician during up-titration. For those patients receiving digitalis, diuretics and ACE inhibitors, dosing of these drugs should be stabilized before initiation of carvedilol treatment.

The recommended dose for initiation of therapy is 3.125 mg twice daily for two weeks. If this dose is tolerated, the dose may thereafter be increased, at intervals of not less than two weeks, to 6.25 mg, 12.5 mg and 25 mg twice daily. Doses should be increased to the highest level tolerated by the patient.

The maximum recommended dose is 25 mg twice daily for all patients with severe CHF and for patients with mild to moderate CHF weighing less than 85 kg (187 lbs). In patients with mild or moderate CHF weighing more than 85 kg, the maximum recommended dose is 50 mg twice daily.

Before each dose increase, the patient should be evaluated by the physician for symptoms of vasodilation or worsening heart failure. Transient worsening of heart failure or fluid retention should be treated with increased doses of diuretics. Occasionally, it may be necessary to lower the dose of carvedilol and, in rare cases, temporarily discontinue carvedilol treatment.

If carvedilol treatment is discontinued for more than one week, therapy should be recommenced at a lower dose level (twice daily) and up-titrated in line with the above dosing recommendation. If carvedilol is discontinued for more than two weeks, therapy should be recommenced at 3.125 mg in line with the above dosing recommendation.

Symptoms of vasodilation may be managed initially by a reduction in the dose of diuretics. If symptoms persist, the dose of ACE inhibitor (if used) may be reduced, followed by a reduction in the dose of carvedilol if necessary. Under these circumstances, the dose of carvedilol should not be increased until symptoms of worsening heart failure or vasodilation have been stabilized.

Contraindications

- hypersensitivity to carvedilol or to any of the excipients
- heart failure belonging to NYHA Class IV of the heart failure classification with marked fluid retention requiring intravenous inotropic treatment
- chronic obstructive pulmonary disease with bronchial obstruction (see *Warnings and Precautions*)
- clinically significant hepatic dysfunction
- bronchial asthma
- second or third-degree AV block (unless a permanent pacemaker is in place)
- severe bradycardia (< 50 beats per minute)
- cardiogenic shock
- sick sinus syndrome (including sinoatrial block)
- severe hypotension (systolic blood pressure below 85 mmHg)
- metabolic acidosis
- Prinzmetal's angina
- untreated phaeochromocytoma
- severe peripheral arterial circulatory disturbances
- Concomitant intravenous treatment with verapamil or diltiazem (see *Drug Interactions*)

Warnings & Precautions

Chronic congestive heart failure

In congestive heart failure patients, worsening cardiac failure or fluid retention may occur during up-titration of carvedilol. If such symptoms occur, diuretics should be increased and the carvedilol dose should not be further increased until clinical stability resumes. Occasionally, it may be necessary to lower the carvedilol dose or, in rare cases, temporarily discontinue it. Such episodes do not preclude subsequent successful titration of carvedilol. Carvedilol should be used with caution in combination with digitalis glycosides, as both medicinal products slow AV conduction (see section interaction with other medicament).

Renal function in congestive heart failure

Reversible deterioration of renal function has been observed with carvedilol therapy in chronic heart failure patients with low blood pressure (systolic BP <100 mmHg), ischaemic heart disease and diffuse vascular disease, and/or underlying renal insufficiency.

Left ventricular dysfunction following acute myocardial infarction

Before treatment with carvedilol is initiated the patient must be clinically stable and should have received an ACE inhibitor for at least the preceding 48 hours, and the dose of the ACE inhibitor should have been stable for at least the preceding 24 hours.

Chronic obstructive pulmonary disease

Carvedilol should be used with caution, in patients with chronic obstructive pulmonary disease (COPD) with a bronchospastic component who are not receiving oral or inhaled medication, and only if the potential benefit outweighs the potential risk. In patients with a tendency to bronchospasm, respiratory distress can occur as a result of a possible increase in airway resistance. Patients should be closely monitored during initiation and up-titration of carvedilol and the dose of carvedilol should be reduced if any evidence of bronchospasm is observed during treatment.

Diabetes

Care should be taken in the administration of carvedilol to patients with diabetes mellitus, as it may be associated with worsening control of blood glucose, or the early signs and symptoms of acute hypoglycaemia may be masked or attenuated.

Peripheral vascular disease and Raynaud's disease

Carvedilol should be used with caution in patients with peripheral vascular disease (e.g., Raynaud's disease) as β -receptor blockers can precipitate or aggravate symptoms of arterial insufficiency.

Thyrotoxicosis

Carvedilol may obscure the symptoms of thyrotoxicosis.

Anaesthesia and major surgery

Caution should be exercised in patients undergoing general surgery, because of the synergistic negative inotropic effects of carvedilol and anaesthetic medicinal products.

Bradycardia

Carvedilol may induce bradycardia. If the patient's pulse rate decreases to less than 55 beats per minute, the dose of carvedilol should be reduced.

Hypersensitivity

Care should be taken in administering carvedilol to patients with a history of serious hypersensitivity reactions, and in patients undergoing desensitisation therapy, as beta-blockers may increase both the sensitivity towards allergens and the severity of hypersensitivity reactions

Severe skin reactions

Very rare cases of severe skin reactions such as toxic epidermal necrolysis (TEN) and Stevens Johnson syndrome (SJS) have been reported during treatment with carvedilol (see also section Side effects). Carvedilol should be permanently discontinued in patients experiencing severe skin reactions attributable to carvedilol.

Psoriasis

Patients with a history of psoriasis associated with beta-blocker therapy should take carvedilol only after consideration of the risk-benefit ratio.

Concomitant use of calcium channel blockers

Careful monitoring of ECG and blood pressure is necessary in patients receiving concomitant therapy with calcium channel blockers of the verapamil or diltiazem type or other antiarrhythmic medicinal products.

Phaeochromocytoma

In patients with phaeochromocytoma, an alpha-blocking agent should be initiated prior to the use of any beta-blocking agent. Although carvedilol has both alpha- and beta-blocking pharmacological activities, there is no experience with its use in this condition. Caution should therefore be taken in the administration of carvedilol to patients suspected of having phaeochromocytoma.

Prinzmetal's variant angina

Agents with non-selective beta-blocking activity may provoke chest pain in patients with Prinzmetal's variant angina. There is no clinical experience with carvedilol in these patients although the alpha-blocking activity of carvedilol may prevent such symptoms. Caution should, however, be taken in the administration of Carvedilol to patients suspected of having Prinzmetal's variant Angina.

Contact lenses

Wearers of contact lenses should bear in mind the possibility of reduced lacrimation.

Withdrawal syndrome

Carvedilol treatment should not be discontinued abruptly, particularly in patients suffering from ischaemic heart disease. The withdrawal of carvedilol should be gradual (over a period of two weeks).

Effects on ability to drive and to use machines

No studies have been performed on the effects of carvedilol on patients' fitness to drive or to operate machinery. Because of individually variable reactions (e.g. dizziness, tiredness), the ability to drive, operate machinery, or work without firm support may be impaired. This applies particularly at the start of treatment, after dose increases, on changing products, and in combination with alcohol.

Drug Interactions

Antiarrhythmics

Isolated cases of conduction disturbances, rarely with haemodynamic disruption, have been observed in patients taking carvedilol and (oral) diltiazem, verapamil and/or amiodarone. As with other beta-blockers, careful monitoring of the ECG and blood pressure should be undertaken when co-administering calcium channel blockers of the verapamil and diltiazem type, as there is the risk of AV conduction disorder or risk of cardiac failure (synergistic effect). Close monitoring should be done in case of co- administration of carvedilol, and class I antiarrhythmics or (oral) amiodarone therapy. Bradycardia, cardiac arrest, and ventricular fibrillation have been reported shortly after initiation of beta-blocker treatment in patients receiving amiodarone. There is a risk of cardiac failure in case of class Ia or Ic antiarrhythmics concomitant intravenous therapy.

Concomitant treatment with reserpine, guanethidine, methyldopa, guanfacine and monoamine oxidase inhibitors (exception MAO-B inhibitors) can lead to an additional decrease in the heart rate and hypotension monitoring of vital signs is recommended.

Dihydropyridines

The administration of dihydropyridines and carvedilol should be done under close supervision, as heart failure and severe hypotension have been reported.

Nitrates

Increased hypotensive effects

Cardiac glycosides

An increase in steady state digoxin levels by approximately 16% has been seen in hypertensive patients in connection with the concomitant use of carvedilol and digoxin. Monitoring of plasma digoxin concentrations is recommended when initiating, discontinuing or adjusting treatment with carvedilol.

Other antihypertensives

Carvedilol may potentiate the effects of other concomitantly administered antihypertensives (e.g. α 1-receptor antagonists) and medicine with antihypertensive adverse reactions such as barbiturates, phenothiazines, tricyclic antidepressants, vasodilators and alcohol.

Ciclosporin

The plasma level of ciclosporin is increased when carvedilol is co-administered. Due to wide interindividual variability in the dose adjustments required, it is recommended that cyclosporine concentrations be monitored closely after initiation of carvedilol therapy and that the dose of cyclosporine be adjusted as appropriate.

Antidiabetics including insulin

The blood sugar-lowering effect of insulin and oral antidiabetics may be intensified. Symptoms of hypoglycaemia may be masked. In diabetic patients, regular monitoring of blood glucose levels is necessary.

Clonidine

When combination treatment with carvedilol and clonidine is discontinued, carvedilol should be withdrawn several days before gradually decreasing the dose of clonidine.

Inhalational anaesthetics

Caution is advised in case of anaesthesia due to synergistic, negative inotrope and hypotensive effect of carvedilol and certain anaesthetics.

NSAIDs, oestrogens and corticosteroids

The antihypertensive effect of carvedilol is decreased due to water and sodium retention.

Medicines inducing or inhibiting cytochrome P450 enzymes

Patients receiving medicines that induce (e.g. rifampicin and barbiturates) or inhibit (e.g. cimetidine, ketoconazole, fluoxetine, haloperidol, verapamil, erythromycin) cytochrome P450 enzymes must closely be monitored during concomitant treatment with carvedilol, as serum carvedilol concentrations may be reduced by the first agents and increased by the enzyme inhibitors.

Rifampicin reduced plasma concentrations of carvedilol by about 70%. Cimetidine increased AUC by about 30% but caused no change in C_{max}. Care may be required in those patients receiving inducers of mixed function oxidases e.g. rifampicin, as serum levels of carvedilol may be reduced, or inhibitors of mixed function oxidases e.g. cimetidine, as serum levels may be increased. However, based on the relatively small effect of cimetidine on carvedilol drug levels, the likelihood of any clinically important interaction is minimal.

Sympathomimetics with alpha-mimetic and beta-mimetic effects

Risk of hypertension and excessive bradycardia.

Ergotamines

Vasoconstriction increased.

Muscle relaxants

Increased neuromuscular block.

Pregnancy and Lactation

Pregnancy

There are no adequate data from the use of carvedilol in pregnant woman. Studies in animals have shown reproductive toxicity. The potential risk for humans is unknown.

Beta-blockers reduce placental perfusion which may result in intrauterine foetal death and immature and premature deliveries. In addition, adverse reactions (especially hypoglycaemia, hypotension, bradycardia, respiratory depression and hypothermia) may occur in the foetus and neonate. There is an increased risk of cardiac and pulmonary complications in the neonate in the postnatal period. Carvedilol should not be used during pregnancy unless clearly necessary (that is if the potential benefit for the mother outweighs the potential risk for the foetus/neonate). The treatment should be stopped 2-3 days before expected birth. If this is not possible the new-born has to be monitored for the first 2-3 days of life.

Breastfeeding

Carvedilol is lipophilic and according to results from studies with lactating animals, carvedilol and its metabolites are excreted in breast milk and therefore mothers receiving carvedilol should not breast-feed.

Side Effects

Infections and infestations

Common: Bronchitis, pneumonia, upper respiratory tract infection, urinary tract infection

Blood and lymphatic system disorders

Common: Anaemia
Rare: Thrombocytopenia
Very rare: Leukopenia

Immune system disorders

Very rare: Hypersensitivity (allergic reaction)

Metabolism and nutrition disorders

Common: Weight increase, hypercholesterolaemia, impaired blood glucose control (hyperglycaemia, hypoglycaemia) in patients with pre-existing diabetes

Psychiatric disorders

Common: Depression, depressed mood
Uncommon: Sleep disorders

Nervous system disorders

Very common: Dizziness, headache
Uncommon: Presyncope, syncope, paraesthesia

Eye disorders

Common: Visual impairment, lacrimation decreased (dry eye), eye irritation

Cardiac disorders

Very common: Cardiac failure

Common: Bradycardia, oedema, hypervolemia, fluid overload
Uncommon: Atrioventricular block, angina pectoris

Vascular disorders

Very common: Hypotension

Common: Orthostatic hypotension, disturbances of peripheral circulation (cold extremities, peripheral vascular disease, exacerbation of intermittent claudication and Reynaud's phenomenon), hypertension

Respiratory, thoracic and mediastinal disorders

Common: Dyspnoea, pulmonary oedema, asthma in predisposed patients
Rare: Nasal congestion

Gastrointestinal disorders

Common: Nausea, diarrhoea, vomiting, dyspepsia, abdominal pain
Uncommon: Constipation
Rare: Dry mouth

Hepatobiliary disorders

Very rare: Alanine aminotransferase (ALT), aspartate aminotransferase (AST) and gamma-glutamyltransferase (GGT) increased

Skin and subcutaneous tissue disorders

Uncommon: Skin reactions (e.g. allergic exanthema, dermatitis, urticaria, pruritus, psoriatic and lichen planus like skin lesions), alopecia
Very rare: Severe cutaneous adverse reactions (e.g. Erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis)

Musculoskeletal and connective tissue disorders

Common: Pain in extremities

Renal and urinary disorders

Common: Renal failure and renal function abnormalities in patients with diffuse vascular disease and/or underlying renal insufficiency, micturition disorders
Very rare: Urinary incontinence in women

Reproductive system and breast disorders

Uncommon: Erectile dysfunction

General disorders and administration site conditions

Very common: Asthenia (fatigue)
Common: Pain, oedema

(c) Description of selected adverse reactions

Dizziness, syncope, headache and asthenia are usually mild and are more likely to occur at the beginning of treatment.

In patients with congestive heart failure, worsening cardiac failure and fluid retention may occur during up-titration of carvedilol dose (see Warnings and Precautions).

Cardiac failure is a commonly reported adverse event in both placebo and carvedilol-treated patients (14.5% and 15.4% respectively, in patients with left ventricular dysfunction following acute myocardial infarction).

Reversible deterioration of renal function has been observed with carvedilol therapy in chronic heart failure patients with low blood pressure, ischaemic heart disease and diffuse vascular disease and/or underlying renal insufficiency (see Warnings & Precautions).

As a class, beta-adrenergic receptor blockers may cause latent diabetes to become manifest, manifest diabetes to be aggravated, and blood glucose counter-regulation to be inhibited.

Carvedilol may cause urinary incontinence in women which resolves upon discontinuation of the medication.

Symptoms and Treatments of Overdose

Symptoms

Overdose may cause serious hypotension, bradycardia, heart failure, cardiogenic shock and cardiac arrest. There may also be respiratory problems, bronchospasm, vomiting, reduced consciousness and convulsions.

Treatment

In addition to normal treatment procedures (such as gastric lavage, pharmacologically induced emesis or activated charcoal), vital signs must be monitored and, if necessary, corrected under intensive care conditions. The following supportive measures may be taken.

Atropine: 0.5 - 2 mg intravenously (for treatment of severe bradycardia).

Glucagon: initially 1 - 10 mg intravenously, followed – if necessary – by slow infusion of 2 - 5 mg/hour (in order to maintain cardiovascular function)

Sympathomimetics according to body-weight and effect: dobutamine, isoprenaline, orciprenaline or adrenaline.

If peripheral vasodilation is the dominant symptom of overdose, the patient has to be given noradrenaline or etilefrine. The patients circulation must be monitored continuously.

If the patient has bradycardia unresponsive to pharmacotherapy, pacemaker therapy should be started.

For the treatment of bronchospasm, the patient should be given beta-sympathomimetics (as aerosol or intravenously if the aerosol does not provide adequate effect) or theophylline intravenously.

If the patient has convulsions, diazepam or clonazepam may be administered as slow intravenous injection.

Carvedilol is highly protein-bound. Therefore, it cannot be eliminated by dialysis.

Important!

In cases of severe overdose with symptoms of shock, supportive treatment should be continued for a sufficiently long period of time, since prolonged elimination and redistribution of carvedilol are to be expected. Duration of the antidote treatment depends on the seriousness of the overdose; supportive treatment must be continued until the patient stabilizes

Method and duration of administration

The tablets do not need to be taken with a meal. However, it is recommended that heart failure patients take their carvedilol medication with food to allow the absorption to be slower and the risk of orthostatic hypotension to be reduced.

Excipients

Lactose monohydrate

Cellulose, microcrystalline
Crospovidone
Povidone K30
Silica, colloidal anhydrous
Magnesium stearate

Carvedilol HEXAL® 6.25mg Tablet: Ferric oxide, Yellow (E172)

Carvedilol HEXAL® 12.5mg Tablet: Ferric oxide, Red (E172)
Ferric oxide, Yellow (E172)

Packing:

Blister pack of 10 tablets. Box of 30, 50 and 100 tablets.

Shelf life: Please refer to the outer box label.

Store in a cool dry place, not above 30°C.

Protect from light.

Keep out of reach of children.

Jauhi daripada kanak-kanak.

Date of Revision: November 2021

Keep medicines out of reach of children

Manufactured for:

Hexal AG

Holzkirchen, Germany.

By:

Salutas Pharma GmbH

(a subsidiary of Hexal AG)

Otto-Von-Guericke Allee 1

D-39179 Barleben

Germany

